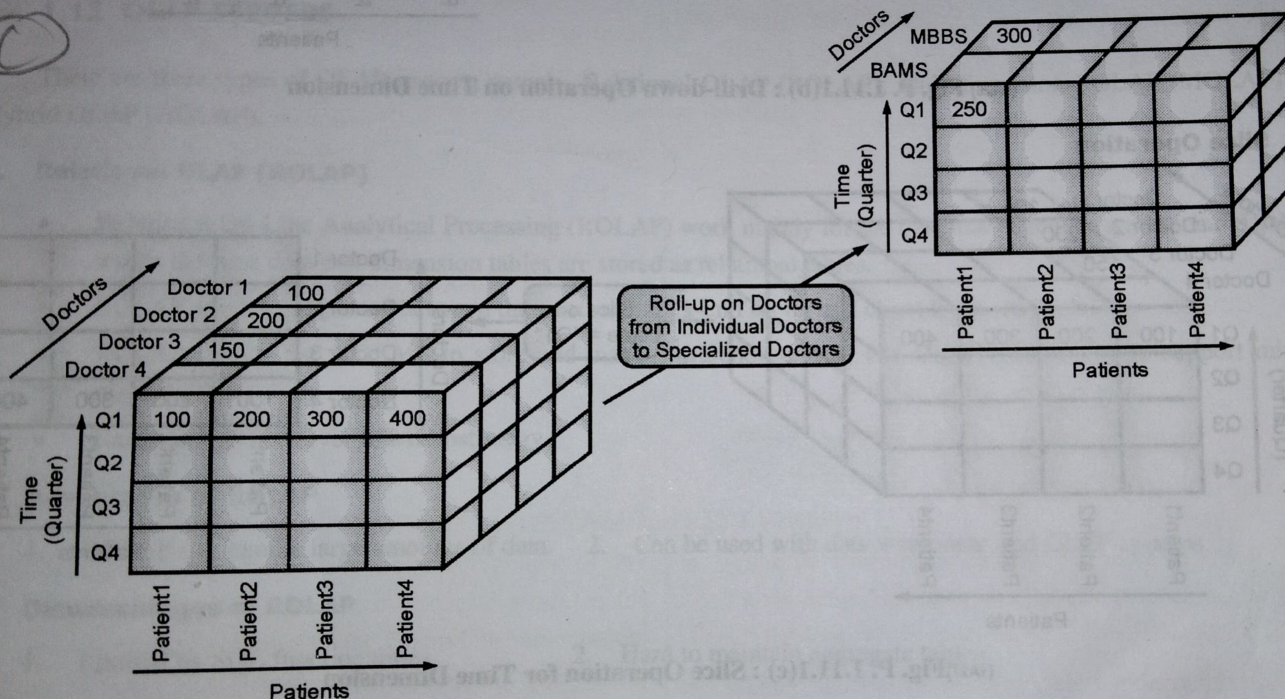
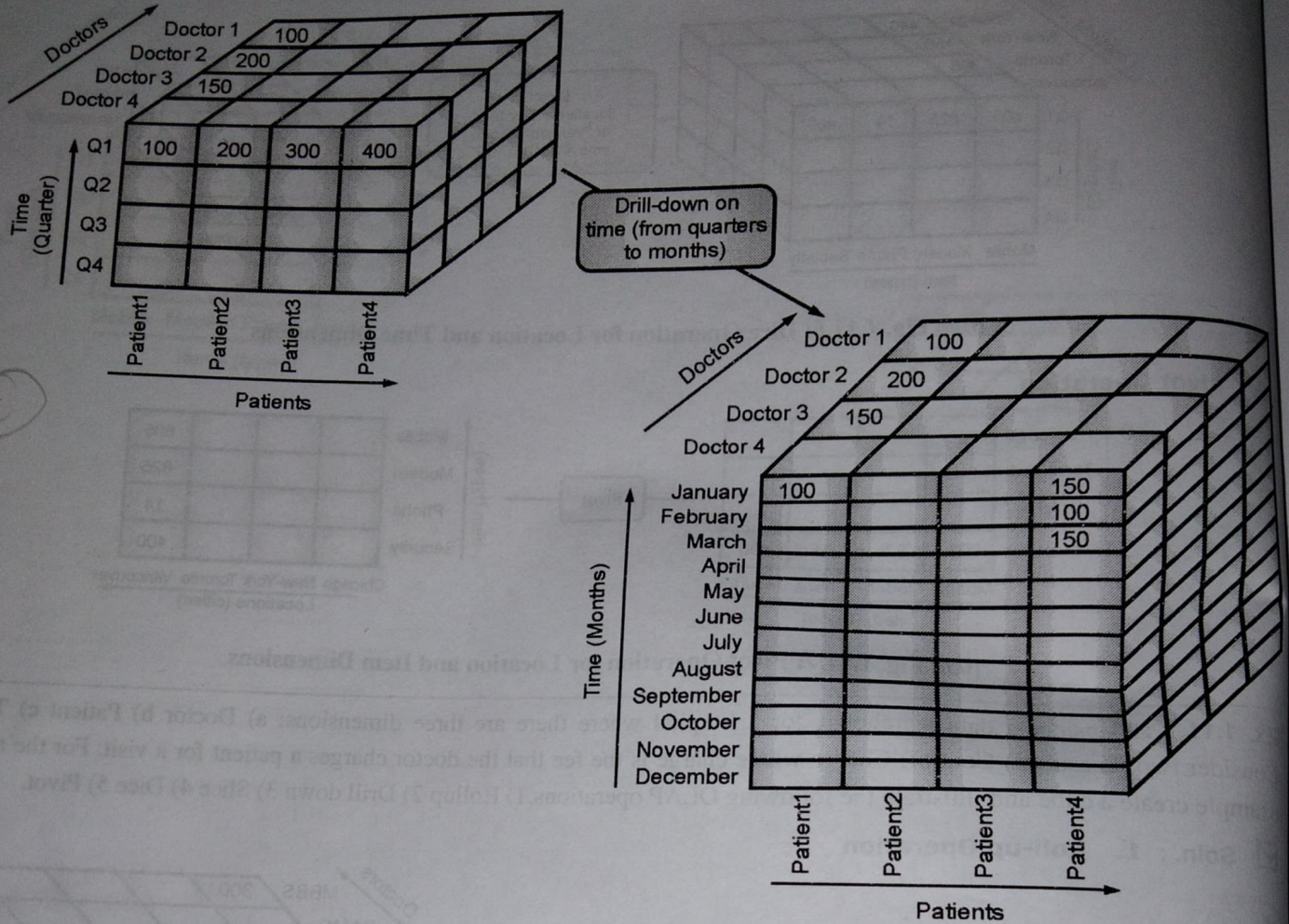


Ex. 1.11.1 : Consider a data warehouse for a hospital where there are three dimensions: a) Doctor b) Patient c) Time. Consider two measures i) Count ii) Charge where charge is the fee that the doctor charges a patient for a visit. For the above example create a cube and illustrate the following OLAP operations. 1) Rollup 2) Drill down 3) Slice 4) Dice 5) Pivot.

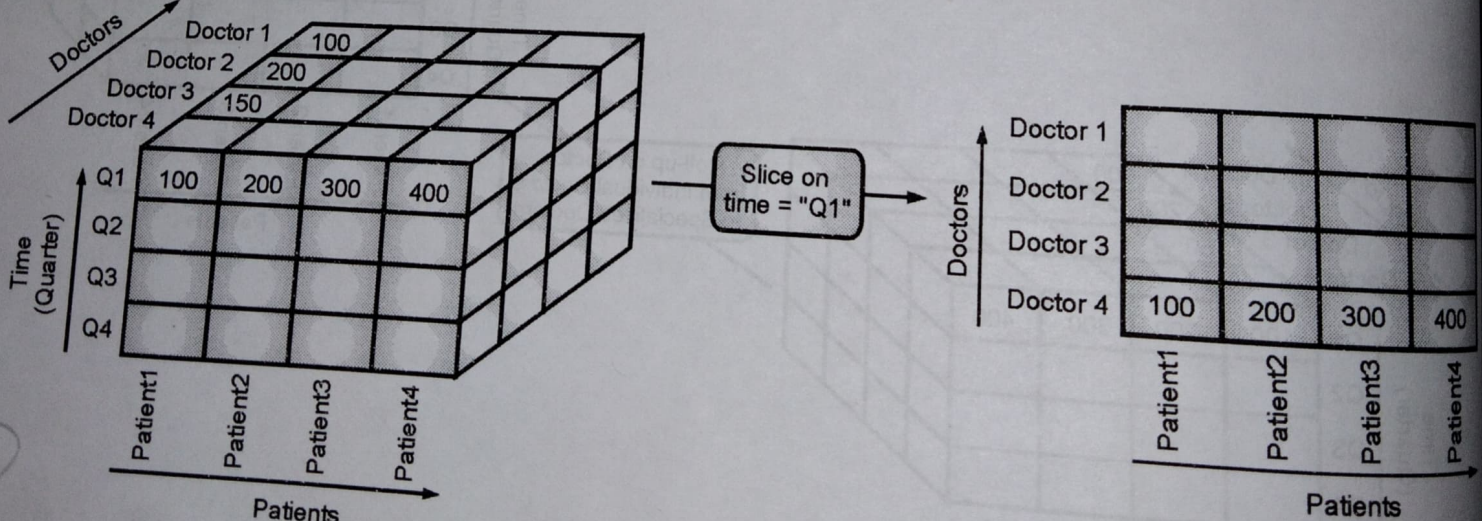
✓ **Soln. : 1. Roll-up Operation**



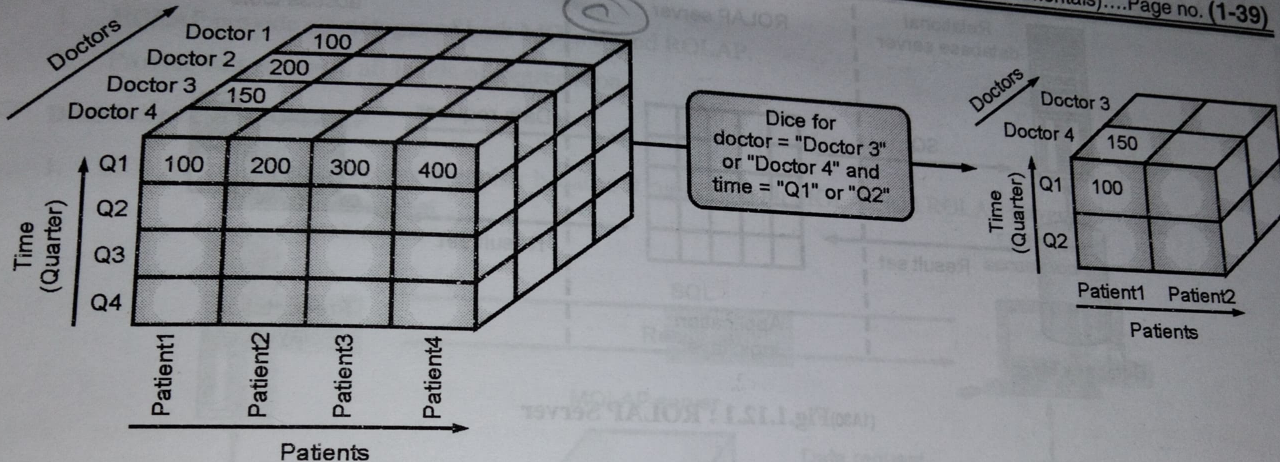
(1A25) Fig. P. 1.11.1(a): Roll-up Operation on Doctors Dimension

2. Drill-down Operation

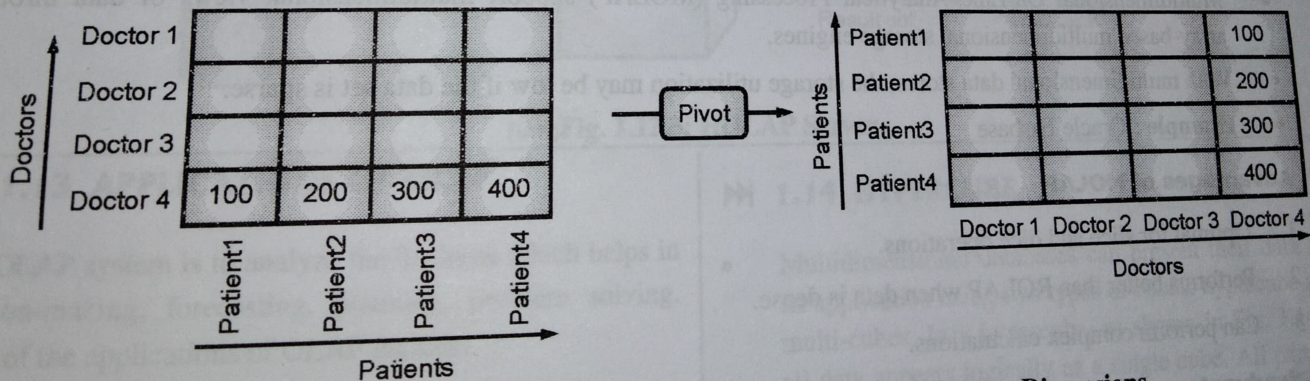
(1A26) Fig. P. 1.11.1(b) : Drill-down Operation on Time Dimension

3. Slice Operation

(1A27) Fig. P. 1.11.1(c) : Slice Operation for Time Dimension

4. Dice Operation

(1A28) Fig. P. 1.11.1(d) : Dice Operation on Doctors and Time Dimensions



(1A29) Fig. P. 1.11.1(e) : Pivot Operation on Doctors and Patients Dimensions

Q7 Foreign Supermarket chain
→ 7 Star Schema

Product
Product-key
Product-Description
Product-Category
Weight
Product-Brand

↑

Time	Sales fact Table	Store
Time-key	Product-key (FK)	Store-key
Date	Time-key (FK)	Store-Name
Day-of-week	Store-key (FK)	Region
Month	Promotion-key (FK)	City
Quarter	Unit-Sales	State
Year	Dollar-Sales	Country
	Dollar-Cost	Zipcode

↓

Promotion-
Promotion-key
Promotion-Name
Promotion-Type
Start-Date
End-Date
Promotion-Cost

$$\begin{array}{r} 365 \\ \times 5 \\ \hline 1825 \end{array}$$

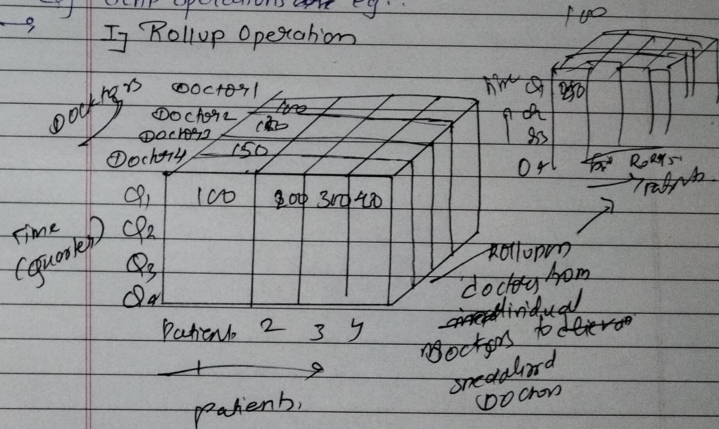
ii) Fact Table records

- Time periods = 5 years = 5×365 days = 1825 days
- No. of stores = 300 stores
- Product in each store daily sell = 4000
- Promotion = 1

$$\begin{aligned} \therefore \text{Maximum no. of fact table records} \\ &= 1825 \times 300 \times 4000 \times 1 \\ &= 219000000 \text{ records} \end{aligned}$$

Q8 OLAP operations are eg.:

→ Rollup Operation



Q7 Compare: Star schema & snowflake schema.

Parameter	<u>Star schema</u>	<u>Snowflake Schema</u>
Ease of Maintenance / change	It has Redundant data & less easy to maintain / change.	It has NO Redundancy and more easy to maintain / change.
Normalization	It has De-normalized tables	It has Normalized tables.
	Star Schema has a central fact table connected directly to dimensional tables.	Snowflake Schema has a central fact table connected to Normalized ^(many) dimension table.
Query Performance	Queries run faster	Queries run slower
Joins	Fewer joins	Higher Joins
Ease of Use.	Simple to design & understand.	more complex to design & understand.
Space.	Star schema uses more storage space.	Snowflake Schema uses less storage space.
Parent table.	No parent table for dimension table.	1 or more Parent table for Dimension table.
When to use.	Best for small or medium warehouse	Better for large warehouse.

Q7 Difference between: OLTP & OLAP operations

Parameters	<u>OLTP</u>	<u>OLAP</u>
name.	OLTP stands for Online Transaction Processing.	OLAP stands for Online Analytical Processing.
Purpose.	OLTP is used to run real time business processing data (day to day).	OLAP is used to run Past time data, analysis of business
Response time.	In Millisecond.	In seconds to Minutes.
Design	DB design is Application Oriented.	DB design is Subject Oriented
Operations	OLTP performs INSERT, UPDATE, DELETE operations.	OLAP performs Roll-up, drill-down, slice, dice & pivot opt.
Handles	OLTP handles many small transactions at a time.	OLAP handles fewer but complex queries.
Storing	OLTP stores Current & detailed data.	OLAP stores Historical & Summarized data.
Query	OLTP provides fast query, processing for routine tasks.	OLAP provides slow query, but deep analysis of data.
Example	Eg. ATM withdrawal, railway booking etc	Eg. Sales trend analysis, profit analysis, etc.

* Answers :-

Q7 Compare : Data Warehouse & Data Mining

Data WarehouseData Mining

i] Data Warehousing is a method of Centralizing data from different sources into one common repository.

ij] Data mining is a method of Comparing large amounts of data to finding right patterns.

ii] Must take place before Data Mining.

ii] After Data Warehousing process

iii] Carried out by Engineers

iii] Business users with help of Engineers.

iv] Data is saved on a regular basis.

iv] Data is analyzed on a regular basis

v] DW. shows what has happened.

v] D.M. explains why & what will happen.

vij] DW use for OLAP operation

vij] Dm used for Algo. prediction & classification.

vii] DW focuses on Storage & management.

vii] D.M. focuses on Analysis and discovery.

viii] To arrange data for easy retrieval, reporting & analysis

viii] To discover knowledge & support decision making.

ix] Eg., stores last 5yr sales.

ix] Eg., stores to Predict last 5yr sales.

Q7 Suppose that the data for analysis includes the attribute salary. We have the foll. values for salary (in thousands of dollars) shown in rising order:

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

i) calculate Mean, Median, Mode & Midrange of data?

ii) find 1st quartile (Q_1) & third quartile (Q_3) of data.

iii) show the boxplot of the data.

→ We have salary data (in thousands of dollars):

30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

i) Mean, Median, mode & midrange?

$$a) \text{ mean} = \frac{x_1 + x_2 + \dots + x_n}{n} \quad (\text{Sum of all values})$$

$$= \frac{696}{12}$$

$$\therefore \text{mean} = 58$$

b) median = mean of middle 2 values of set

$$= (52 + 56)$$

$$\therefore \text{median} = 54$$

c) mode (most frequent values)

here 52, 70. both appear twice.

$$\therefore \text{Mode} = 52, 70$$

$$d) \text{ midrange} = \frac{(\text{min} + \text{max})}{2}$$

$$= \frac{30 + 110}{2}$$

$$\therefore \text{median} = 70$$

ii) Quartiles

$\therefore Q_1 = \text{Median of lower half}$

$$= \frac{47 + 50}{2}$$

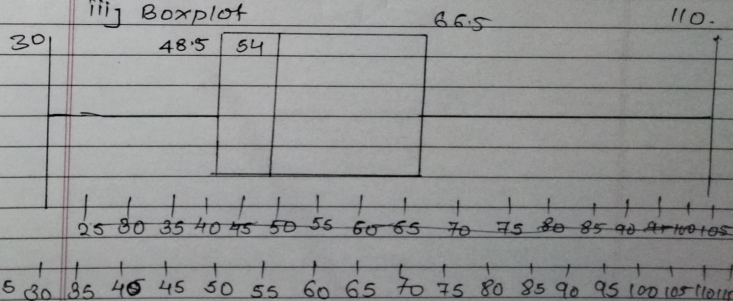
$$\therefore Q_1 = 48.5$$

$\therefore Q_3 = \text{Median of Upper half}$

$$= \frac{63 + 70}{2}$$

$$\therefore Q_3 = 66.5$$

iii) Boxplot



Q. Describe in details Data warehouse also explain its features, needs, appl'n, benefits & approaches.

→ An Data Warehouse (DW) is a centralized, integrated repository of historical and current data & collection from multiple sources, organized to support decision-making, reporting and data analysis.

Features of Data Warehouse :-

- 1) Subject Oriented: organized around major business
- 2) It is Integrated: Data is collected from diff. sources and made consistent.
- 3) Time Variant: Stores historical data.
- 4) It is Non-volatile: Once entered into Warehouse the data should not be change.

Needs of Data Warehouse :-

- 1) Business User: ~~Business~~ Business users ~~can~~ require a data warehouse to view summarize^d data from past.
- 2) Stores historical data: DW is req. to store the time variable data from past.
- 3) Better decision making:
- 4) For data consistency & quality

Appln of Data Warehouse :-

- 1) Banking: Widely used in this sector for management of resources available on desk.
- 2) Healthcare: Patient care analysis & disease trend tracking.
- 3) Telecommunications: For product promotions.
- 4) Investment & Insurance sectors:
DW is primarily used to analyze data patterns, customer trends.
- 5) Airline: This is used for operation purpose like crew assignment, etc.

Benefit of DW :-

- 1) Deliver very enhance business intelligence
- 2) Save time
- 3) Enhance data quality and consistency.
- 4) Improve decision making process.

Approaches of Building DW :-

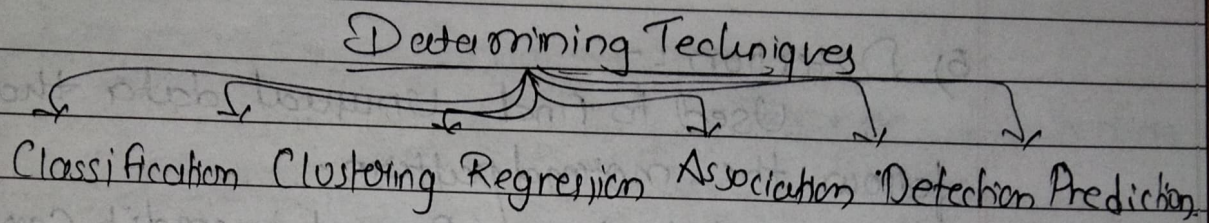
- 1) Top-down approach
- 2) Bottom-up approach
- 3) Hybrid Approach

Q7

Describe in details Data Mining Techniques.

Data mining is the process of discovering the patterns trends and useful information from large data sets.

Different techniques are used depending on the type of data and the goal.



1) Classification :-

Data can be divided into predefined groups or classes.

For eg, emails can be classified as spams or not spams.

2) Clustering :-

Clustering is very similar to Classification.

Similar data objects are grouped together without predefined labels.

eg, ~~customers~~ customers can be grouped based on their buying behaviour.

3) Regression :-

One data keeps relation between 2nd data but it is not necessary to. One data is depending on 2nd and 2nd also depending on 1st. These techniques predict continuous values.

4) Association Rule mining:-

It finds relationships betn items in a data set. ; more specifically to dependently linked variables.

foreg., if customer buys bread, they are also likely to buy butter.

5) Detection:-

Used to find Unusual data that does not follow the normal pattern.

foreg., detecting fraud in credit card transaction.

6) Prediction:-

past data is used to forecast future outcomes. foreg., predicted next month sales.

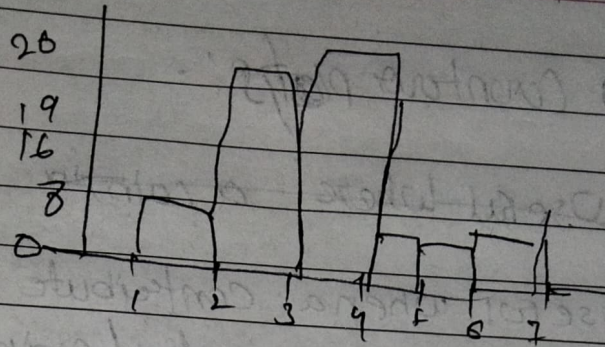
Q9) Explain different data visualization techniques.

Data visualization is the process of representing data in a graphical or visual format so that ~~clear~~ patterns, trends and insights can be easily understood.

Visualization Techniques:-

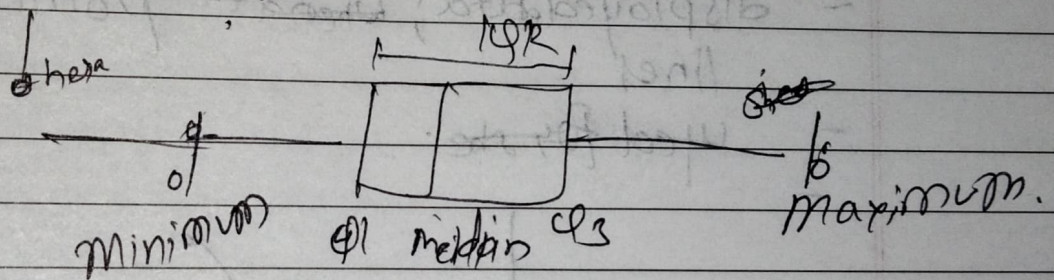
1) Histogram:-

- usually shows the distribution of values of a single variable.
- eg., petal width.



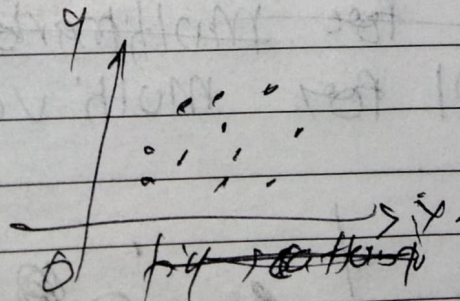
2) Box plots:-

- Invented by J. Tukey
- Another way of displaying the distribution of a data.
- Use to compare ~~attributes~~ attributes.



3) Scatter plot:-

- Attributes values determine the position.
- 2D and 3D both available



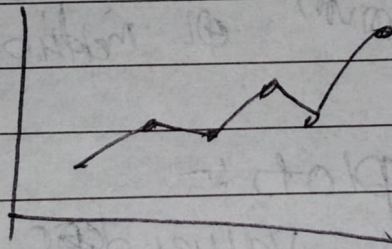
Scatter plot

4) Counter plots:-

- Useful where a continuous attribute is measured on a spatial grid.
- eg.) Can also plot temp., rainfall, air pressure.

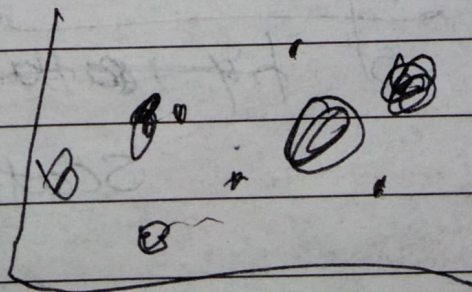
5) Line chart:-

- displayed data; where points connected by lines
- used for time.



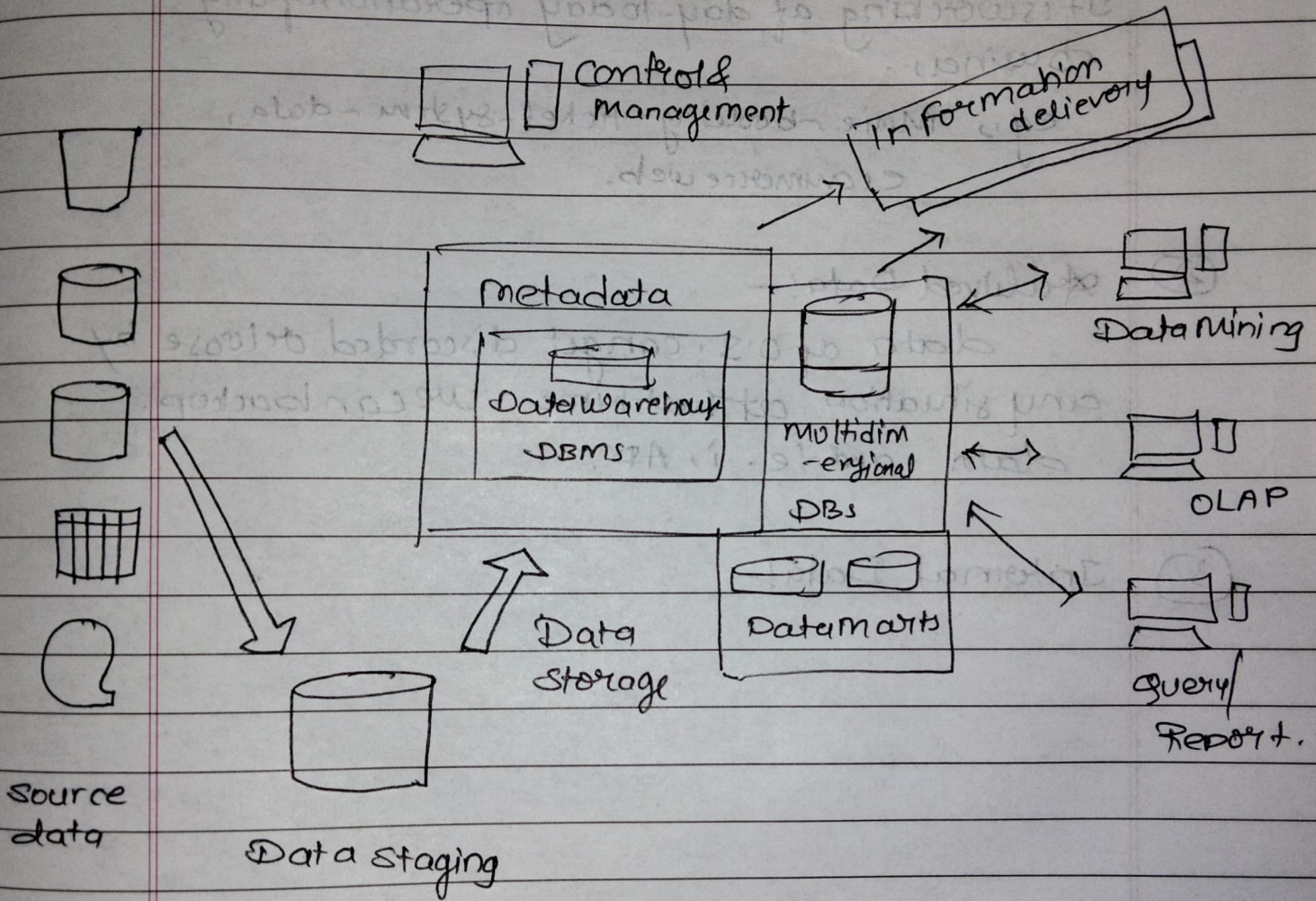
6) Bubble chart

- Useful for Multivariate data.
- Useful for Multi variable comparison

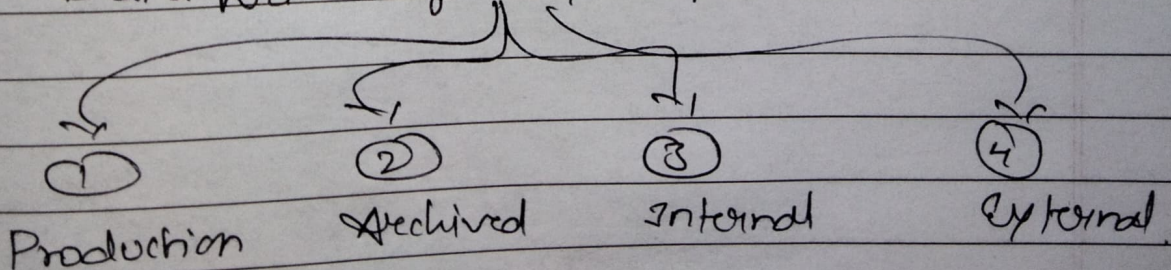


Q7 Data Warehouse Architecture.

- Data Warehouse where data is centralized and stored in one place with organized manner.
- for eg., business stored data, customer data, etc.



- Data Warehouse 4 Types of data



Data Warehouse ET Mining.

Data Warehouse ^{centralized repository}
- where data is stored in one place with organized manner
for eg. A business stored data in form of web, records of sales, Customer data, etc.

Q2. Data Warehouse Architecture.

Defⁿ - of Data Warehouse

Data Warehouse have 3 types of Data.

1. Production data
2. Archived data
3. Internal data
4. External data

1. Production Data.

The Data which is coming from operational Data. Operational Data is Data which is coming from many transactions. for eg. In a bank many transaction happens at same time, and many customer do transaction that type of data is operation system. It is working on day-to-day operation of any business e.g. for bank, e-commerce web.

2. Archived data.

The Data of operational system can get discarded or loose by any situation at that time we can backup the data and that type of data is called as Archived data.

3. Internal Data.

In this type of Data. In company many employee saved the data in private sheet, document, Documental Database and in future most of the cases that data use in Data warehouse and the data is stored internally that's why it is called as Internal Data.

4. External Data.

In this type of Data the data is send to company externally for e.g. In a college a Data is send by external by University to college that data is called as external Data.

Data Staging.

The Data Sources are stored in Staging for temporary stored in purpose. The Data Sources Extracted and send to Data staging for storing Data temporary.

Loading.

The data which is stored in Data staging is start loaded into Data Storage. Loading have 2 types . 1. Initial Loading
2. Incremental Loading

Initial loading → Data loaded Initially . e.g company Data After prod.
Incremental loading → Data loaded one by one additionally
→ any new Data periodically at time on time is called incremental.

EDW Data Storage.

Enterprise Organized Data with the help of multidimensional Data model

Data Warehouse.
It have Meta-Data → which piece of data will go in Data Warehouse.
Loading is happening because of meta data.

Datamarts → From EDW the data is divided into marts for various type of data. It is separated in marts because in EDW there are many data store in it. And because of Datamarts it is easy to fetch data.

Information Delivery

From Data Marts it convert the data in form various type which user wants.

Control & Management.

It control & manage the whole process & interaction is control & manage by it.

Source

Data Mining

OLAP

Report/Query
etc...

Information
Delivery

Users
&
Tools

* Introduction

* Data Sources

- Production data
- Archived data
- Internal data
- External Data

* Data Staging

* Loading

- Initial Loading
- Incremental Loading

* EDW

* Meta Data

* Data Marts

* Information delivery

* Control & Management